

General distinctiveness as an estimate of variation

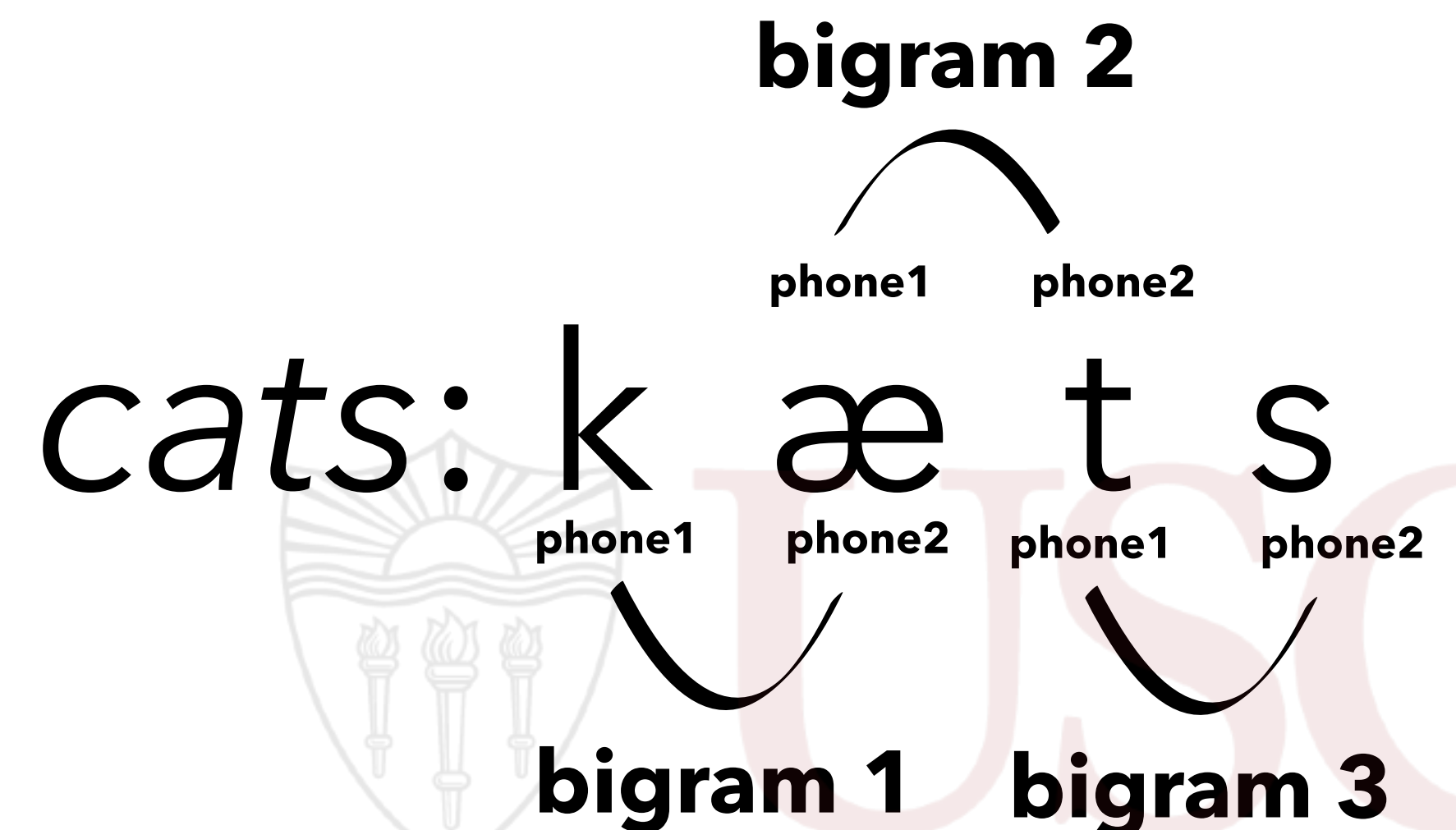
- To investigate the effects of information-theoretic constraints on speech variation in an unsupervised way:
 - How does the contextual information content (**surprisal**) of the bigram, given its previous context in the word, affect the phone2's distinctiveness to phone1?

$$I(\text{bigram} \mid \text{context}) = -\log(P(\text{bigram} \mid \text{context}))$$

$$I(\mathbf{k}\mathbf{\ae} \mid \# __) = -\log(P(\mathbf{k}\mathbf{\ae} \mid \# __))$$

$$I(\mathbf{\ae}\mathbf{t} \mid \mathbf{k} __) = -\log(P(\mathbf{\ae}\mathbf{t} \mid \mathbf{k} __))$$

$$I(\mathbf{t}\mathbf{s} \mid \mathbf{k}\mathbf{\ae} __) = -\log(P(\mathbf{t}\mathbf{s} \mid \mathbf{k}\mathbf{\ae} __))$$



Perceptual recoverability, articulatory efficiency & speech flexibility

- Would information-theoretic constraints behave differently in the articulatory & acoustic domains?